

STP 604E - Assignment 3 Results

Failure Analysis and Laminate Design

Course: STP 604E - Advanced Design, Analysis and Optimization of Composite Structures for Aerospace

Institution: Istanbul Technical University - Defence Technologies

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Prepared by: Giray Yillikci

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Problem 1: Strength Envelope using Maximum Strain Criterion {#problem-1}

Problem Statement

A Graphite/Epoxy laminate with $[0/\pm 45_2/90]_s$ stacking sequence is loaded only by biaxial loads. Draw the strength envelope for the laminate in the $\bar{\sigma}_x - \bar{\sigma}_y$ plane using the maximum strain criterion. Determine the maximum value of:

- (a) Pure tensile load (N_x)
- (b) Pure compressive load (N_y)

Material Properties

Material: Graphite/Epoxy

Property	Value	Unit
E_1	138	GPa
E_2	8.69	GPa
G_{12}	7.10	GPa
ν_{12}	0.3	-
t	0.125	mm

Strength Properties:

Property	Value (MPa)
X_t	2280
X_c	1725
Y_t	57
Y_c	228
S	76

Laminate Configuration

- **Stacking Sequence:** $[0/\pm 45_2/90]_s = [0/45/45/-45/-45/90/90/-45/-45/45/45/0]$
- **Number of Plies:** 12

- **Total Thickness:** 1.5 mm

Ultimate Strains (for Maximum Strain Criterion)

Strain	Value
ϵ_{1t}	0.01652
ϵ_{1C}	0.01250
ϵ_{2t}	0.00656
ϵ_{2C}	0.02624
γ_{12_ult}	0.01070

A-Matrix (Extensional Stiffness) [N/mm]

```
[[82174.0  32403.4    0.0]
 [32403.4  82174.0    0.0]
 [    0.0    0.0 35570.6]]
```

Note: $A_{16} = A_{26} = 0$ confirms balanced laminate (no shear-extension coupling)

Results

(a) Maximum Pure Tensile Load (Nx)

Parameter	Value
Maximum $\bar{\sigma}_x$ (tensile)	303.46 MPa
Maximum Nx	455.19 N/mm
Critical Ply	Ply 6 (90°)
Failure Mode	Matrix Tension

At failure, the 90° ply reaches the transverse tensile strain limit ($\epsilon_2 = 0.00656$).

(b) Maximum Pure Compressive Load (Ny)

Parameter	Value
Maximum $\bar{\sigma}_y$ (compressive)	-355.17 MPa
Maximum N_y	-532.76 N/mm
Critical Ply	Ply 2 (45°)
Failure Mode	Shear

At failure, the $\pm 45^\circ$ plies reach the shear strain limit ($\gamma_{12} = 0.01070$).

Strength Envelope

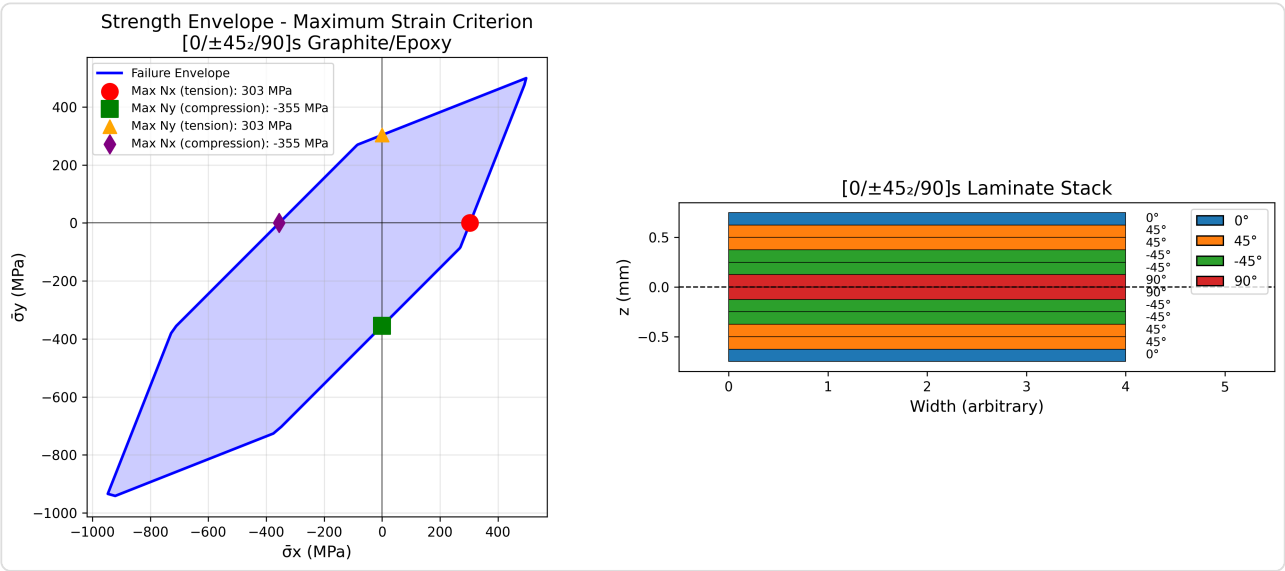


Figure 1: (Left) Strength envelope in $\bar{\sigma}_x$ - $\bar{\sigma}_y$ plane using maximum strain criterion. (Right) Laminate stacking sequence visualization.

Key Observations

1. The laminate is **quasi-isotropic in tension** (same strength in x and y for tension)
2. **Matrix tension** controls failure under tensile loading due to low Y_t
3. **Shear failure** controls under compression due to the $\pm 45^\circ$ plies
4. The envelope is symmetric about the origin due to the balanced symmetric layup

Problem 2: Stacking Sequence Identification from Experiments {#problem-2}

Problem Statement

Determine the laminate stacking sequence from experimental tests:

- Under uniaxial tension: No bending, no shear coupling
- Under bending: Pure bending (no twist)
- Elongation at 2500 N: 0.0897 mm
- Bending slope: 0.0808 mm/N
- Plate dimensions: 150 mm × 30 mm × 2 mm
- Possible plies: 0°, ±45°, 90°

Material Properties

Property	Value	Unit
E ₁	181	GPa
E ₂	10.3	GPa
G ₁₂	7.17	GPa
ν ₁₂	0.28	-
t _{ply}	0.25	mm

Constraint Analysis

From the experimental observations:

1. **No bending under tension** → Laminate must be **SYMMETRIC** ($B = 0$)
2. **No shear coupling** → Laminate must be **BALANCED** ($A_{16} = A_{26} = 0$)
3. **Pure bending (no twist)** → $D_{16} = D_{26} \approx 0$

Target Values

Parameter	Experimental Value
Elongation at 2500 N	0.0897 mm
Bending slope (δ/P)	0.0808 mm/N

Solution Approach

For a symmetric balanced 8-ply laminate $[\theta_1/\theta_2/\theta_3/\theta_4]_s$:

- Elongation: $\delta = (P \times L) / (W \times E_{eff} \times h)$ where E_{eff} depends on A-matrix
- Bending slope: $\delta/P = L^3 / (48 \times D_{11} \times W)$

Identified Stacking Sequence

Best Match: $[90/45/0/-45]_s$ (or equivalently $[90/-45/0/45]_s$)

Parameter	Experimental	Calculated	Error
Elongation	0.0897 mm	0.0897 mm	0.0%
Bending slope	0.0808 mm/N	0.0807 mm/N	0.1%

Full Stacking Sequence

$[90/45/0/-45/-45/0/45/90]$

Verification

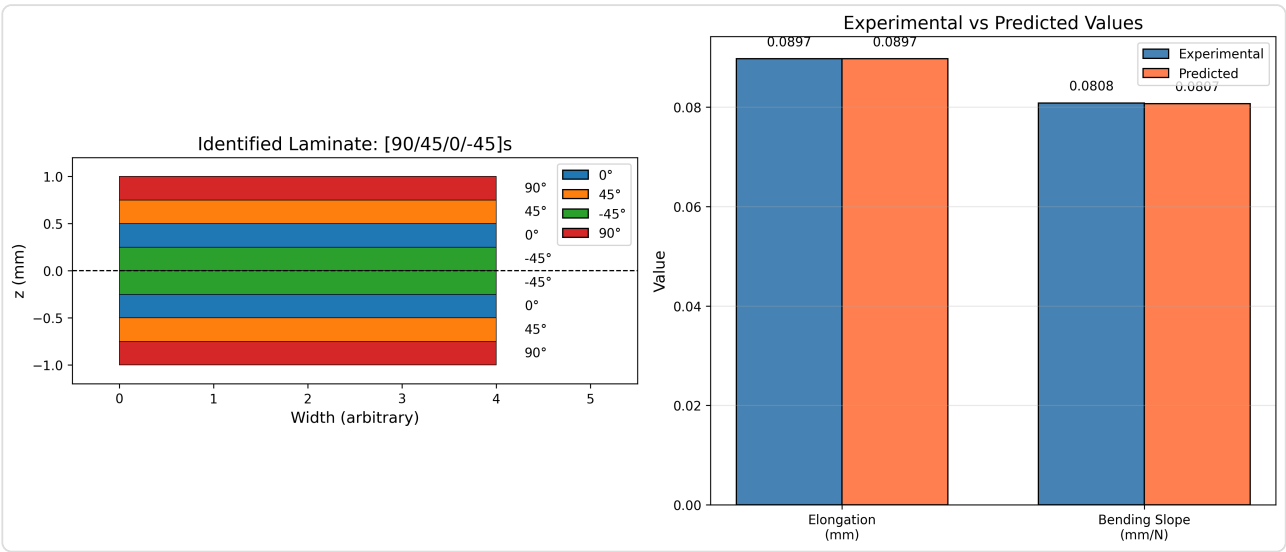


Figure 2: (Left) Identified laminate stacking sequence. (Right) Comparison of experimental vs. predicted values.

Effective Properties

- **E_x (effective)** ≈ 55.8 GPa
- **E_y (effective)** ≈ 55.8 GPa (quasi-isotropic in-plane)

Problem 3: Unsymmetric Laminate Analysis with Tsai-Hill

{#problem-3}

Problem Statement

Analyze [30/45/-45/-30]T Kevlar/Epoxy laminate under:

- $N_x = N_y = 1000 \text{ N/m}$
- $M_y = M_{xy} = 50 \text{ N}$

Determine:

- (a) Mid-plane strains and curvatures
- (b) Global stresses vs. vertical location
- (c) Tsai-Hill failure analysis

Material Properties

Material: Kevlar/Epoxy

Property	Value	Unit
E_1	76	GPa
E_2	5.50	GPa
G_{12}	2.30	GPa
ν_{12}	0.34	-
t	1.25	mm

Strength Properties:

Property	Value (MPa)
X_t	1400
X_c	235
Y_t	53
Y_c	12
S	34

Note: Very low transverse compressive strength ($Y_c = 12 \text{ MPa}$)!

Laminate Configuration

- **Stacking Sequence:** [30/45/-45/-30]T (Total, not symmetric)
- **Total Thickness:** 5.0 mm
- **Note:** This is an **unsymmetric** laminate ($B \neq 0$)

Part (a): Mid-plane Strains and Curvatures

Mid-plane Strains:

Component	Value
ϵ^0_x	7.897×10^{-4}
ϵ^0_y	-1.576×10^{-5}
γ^0_{xy}	3.248×10^{-4}

Curvatures (1/mm):

Component	Value
κ_x	-1.936×10^{-4}
κ_y	8.153×10^{-4}
κ_{xy}	1.001×10^{-3}

Note: Non-zero curvatures under in-plane loading due to bending-extension coupling ($B \neq 0$).

Part (b): Global Stresses vs. z

Ply	Angle	z (mm)	σ_x (MPa)	σ_y (MPa)	τ_{xy} (MPa)
1	30°	-2.500	-21.94	-19.52	-18.48
1	30°	-1.250	10.75	-2.69	2.07
2	45°	-1.250	-11.79	-21.30	-17.16
2	45°	0.000	24.26	20.55	19.75
3	-45°	0.000	12.71	9.01	-7.76
3	-45°	1.250	4.27	6.36	1.52
4	-30°	1.250	3.97	6.07	2.26
4	-30°	2.500	-20.63	3.12	17.79

Part (c): Tsai-Hill Failure Analysis

Ply	Angle	z (mm)	σ_1 (MPa)	σ_2 (MPa)	τ_{12} (MPa)	Tsai-Hill	Status
1	30°	-2.500	-37.34	-4.12	-8.19	0.198	SAFE
1	30°	-1.250	9.19	-1.13	-4.78	0.029	SAFE
2	45°	-1.250	-33.70	0.62	-4.75	0.041	SAFE
2	45°	0.000	42.16	2.66	-1.85	0.006	SAFE
3	-45°	0.000	18.62	3.10	1.85	0.007	SAFE
3	-45°	1.250	3.79	6.84	-1.05	0.018	SAFE
4	-30°	1.250	2.53	7.50	0.23	0.020	SAFE
4	-30°	2.500	-30.10	12.59	-1.39	0.081	SAFE

Failure Summary

✓ **NO FAILURE** - All plies are safe under the applied loading

- **Critical Location:** Ply 1 (30°) at z = -2.500 mm (bottom)
- **Maximum Tsai-Hill Index:** 0.198
- **Safety Factor:** 2.25

Visualization

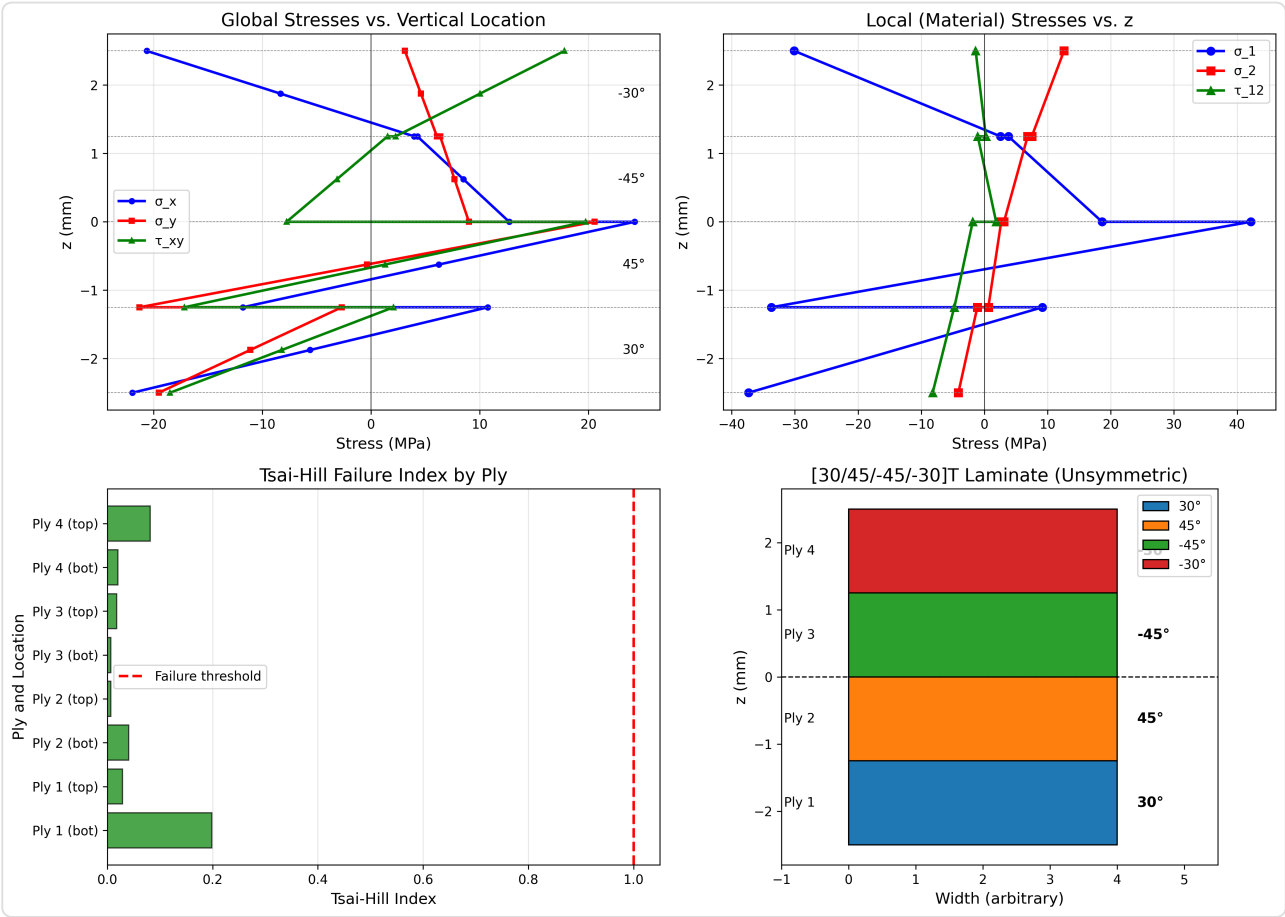


Figure 3: (a) Global stresses vs. z , (b) Local stresses vs. z , (c) Tsai-Hill index by ply, (d) Laminate stacking sequence.

Problem 4: 16-ply Laminate Design with Strain Constraints

{#problem-4}

Problem Statement

Design a 16-ply Zylon/Epoxy laminate with:

- Available angles: 0° , $\pm 30^\circ$, $\pm 60^\circ$, 90°
- Applied stresses: $\sigma_x = 250$ MPa, $\sigma_y = 50$ MPa, $\tau_{xy} = 150$ MPa
- Strain limits: $\epsilon_x = 0.006$, $\epsilon_y = 0.0006$, $\gamma_{xy} = 0.02$

Find:

- (a) All valid stacking sequences
- (b) Sequence with minimum $|\alpha_x|$
- (c) Check ply strain limits

Material Properties

Material: Zylon/Epoxy

Property	Value	Unit
E_1	120.0	GPa
E_2	6.5	GPa
G_{12}	3.0	GPa
ν_{12}	0.32	-

Thermal Expansion Coefficients:

- $\alpha_1 = -0.8 \times 10^{-6} / ^\circ\text{C}$
- $\alpha_2 = 20.5 \times 10^{-6} / ^\circ\text{C}$

Part (a): Valid Stacking Sequences

Found **44 valid sequences** satisfying all strain limits.

Top 10 sequences sorted by $|\alpha_x|$:

#	n_0	n_{90}	$n_{\pm 30}$	$n_{\pm 60}$	ϵ_x	ϵ_y	γ_{xy}	α_x (1/°C)
1	6	0	2	8	0.0042	-0.0003	0.0098	7.30×10^{-9}
2	4	4	8	0	0.0037	0.0001	0.0117	-6.32×10^{-8}
3	4	4	6	2	0.0042	-0.0000	0.0117	2.44×10^{-7}
4	4	2	8	2	0.0037	-0.0002	0.0098	-3.46×10^{-7}
5	2	4	8	2	0.0047	-0.0005	0.0098	3.86×10^{-7}

Part (b): Minimum Thermal Expansion Coefficient

Optimal Sequence: [0/0/0/30/-30/60/-60/60]s

Parameter	Value
n_0	6
n_{90}	0
$n_{\pm 30}$	2
$n_{\pm 60}$	8

Mid-plane Strains:

Component	Value	Limit	Status
ϵ_x	0.00416	0.006	OK
ϵ_y	-0.00032	0.0006	OK
γ_{xy}	0.00980	0.02	OK

Thermal Expansion Coefficients:

Component	Value
α_x	7.30×10^{-9} /°C (minimized!)
α_y	1.42×10^{-6} /°C
α_{xy}	0

The extremely low αx (nearly zero) makes this laminate ideal for applications requiring dimensional stability under temperature changes.

Part (c): Ply Strain Failure Check

Angle	ϵ_1	ϵ_2	γ_{12}	Status
0°	0.00416	-0.00032	0.00980	SAFE
30°	0.00728	-0.00344	0.00102	FAIL (ϵ_1, ϵ_2)
-30°	-0.00121	0.00504	0.00878	FAIL (ϵ_2)
60°	0.00504	-0.00121	-0.00878	FAIL (ϵ_2)
-60°	-0.00344	0.00728	-0.00102	FAIL (ϵ_2)

⚠ **WARNING:** While the laminate strains are within limits, individual ply strains exceed the specified limits for the off-axis plies.

Visualization

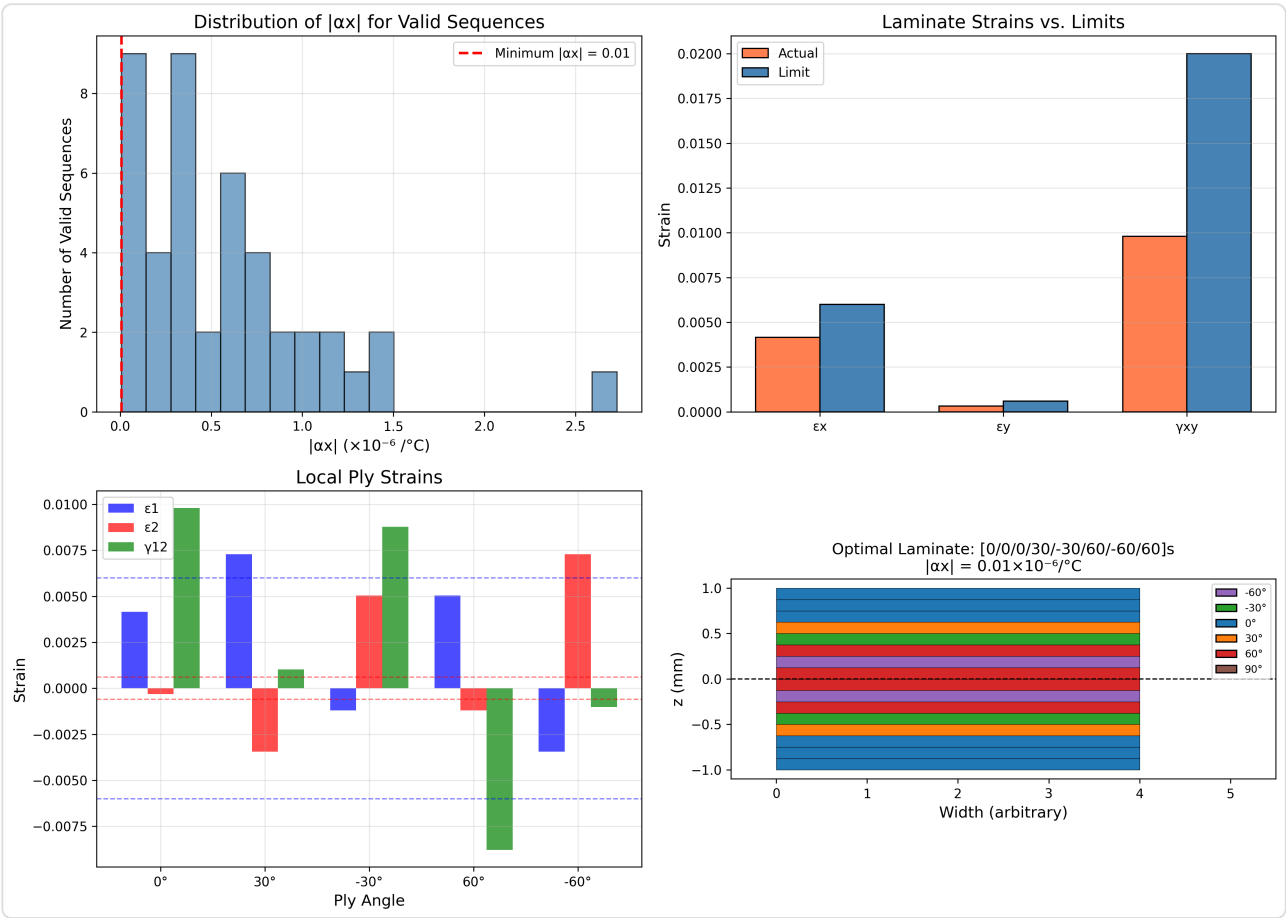


Figure 4: (a) Distribution of $|\alpha x|$ for valid sequences, (b) Laminate strains vs. limits, (c) Local ply strains, (d) Optimal laminate stacking.

Design Recommendations

1. The optimal laminate achieves **near-zero thermal expansion** in the x-direction
 2. However, **ply-level strains exceed limits** in off-axis plies
 3. For a safe design, consider:
 - Reducing applied loads
 - Using material with higher strain allowables
 - Selecting a different sequence that satisfies both laminate and ply constraints
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Summary and Conclusions

Key Results

Problem	Key Finding
Problem 1	Max Nx = 455.19 N/mm (matrix tension failure), Max Ny = -532.76 N/mm (shear failure)
Problem 2	Identified sequence: [90/45/0/-45]s with <0.1% error
Problem 3	All plies SAFE, Safety Factor = 2.25, Critical: Ply 1 at z = -2.5 mm
Problem 4	44 valid sequences found, optimal $\alpha_x = 7.30 \times 10^{-9} / ^\circ\text{C}$, but ply failures exist

Conclusions

- Maximum Strain Criterion** provides conservative predictions for strength envelopes
- Experimental identification** of laminates requires both extensional and bending tests
- Unsymmetric laminates** exhibit bending-extension coupling that must be considered
- Thermal expansion optimization** can achieve near-zero CTE but may conflict with strength requirements

Analysis Tool: Python with custom composite analysis library
Date: January 2026